

VRMLiveViewer AutoCamera JSON Specification

Unofficial reference for generating and refining camera work from natural-language instructions

Document Version: 1.1

This unofficial reference summarizes the observed relationship between JSON saved by VRM Live Viewer Auto Camera Settings and the corresponding UI controls. It is intended to help generative AI translate natural-language camera directions into AutoCamera settings and explain later refinements.

1. Basic JSON structure

```
{ "dataType": "AutoCamera", "mode": "Follow", "param": { "keys": [...], "values": [...] }, "datas": [ { "keys": [...], "values": [...] }, "etc": { "keys": [...], "values": [...] }, "version": "3.14.1" }
```

Important: within each object, keys[n] corresponds to values[n]. Generation and editing must preserve the order and element count of both arrays.

2. Frames and common settings

UI / Function	JSON key	Observed mapping
Start Frame	startFrame	Start frame of the camera data; frame-based and affected by fps
Switching Mode	switchingMode	Frame observed
FPS Unit	fps	30 and 60 verified in tests; affects conversion for startFrame, freezeFrame, etc.
Far Clip Plane	farClip	Corresponds to UI Clip Distance. Verified minimum: 150. Unit, maximum, and real-world
Max Chara No	maxCharaIndex	-1 observed for (All)

3. Basic camera settings

UI / Function	JSON key	Observed mapping
Offset X/Y/Z	offset	X,Y,Z
Distance	distance	Base camera distance
Rotation X/Y/Z	angle	X,Y,Z
VAngle	viewAngle	View angle
Fix Position	fixed	Exact encoding not established
Position	position	Observed in saved JSON; exact UI role not fully established
Speed	speed	Observed in saved JSON; exact UI mapping not fully established

4. Animated/change settings

UI / Function	JSON key	Observed mapping
Changing Position: enabled	enablePositionChange	True/False
Position Interpolation	positionInterp	Linear / EaseInOut observed
One Way (Position)	positionHalf	Likely related; exact boolean semantics not fully established
Add X/Y/Z	positionMax	X,Y,Z
Changing Time	positionTime	Stored value; do not assume the UI value is frames
Timing Offset	positionTimingShift	Stored value
Changing Rotation: enabled	enableRotationChange	True/False
Mode: Min and Max Angle	rotationMode	MinMax

UI / Function	JSON key	Observed mapping
Rotation Interpolation	rotationInterp	Linear / EaseInOut / EaseIn observed. Exact curves and the strength of (low) a
One Way (Rotation)	rotationHalf	True = one way; False = back-and-forth (verified in actual testing)
Min Angle X/Y/Z	rotationMin	X,Y,Z
Max Angle X/Y/Z	rotationMax	X,Y,Z
Changing Time	rotationTime	Time-setting stored value. rotationTime=24 was verified as 12 seconds at both
Timing Offset	rotationTimingShift	Stored value
Changing Distance: enabled	enableDistanceChange	True/False
Distance Interpolation	distanceInterp	Interpolation string
One Way (Distance)	distanceHalf	Likely related
Add Distance	distanceMax	Example zoom-in uses -1.0; example zoom-out uses +1.5
Changing Time	distanceTime	Stored value
Timing Offset	distanceTimingShift	Stored value
Changing VAngle: enabled	enableVAngleChange	True/False
VAngle Interpolation	vangleInterp	Interpolation string
One Way (VAngle)	vangleHalf	Likely related
Max VAngle	vangleMax	Stored value
Changing Time	vangleTime	Stored value
Timing Offset	vangleTimingShift	Stored value

5. Jitter

UI / Function	JSON key	Observed mapping
Jitter enabled	enableJitter	True/False
Position Amount	jitterPosAmount	Position shake amount
Position Freq	jitterPosFreq	Position shake frequency setting
Position Axis X/Y/Z	jitterPosAxisRate	X,Y,Z
Rotation Amount	jitterRotAmount	Rotation shake amount
Rotation Freq	jitterRotFreq	Rotation shake frequency setting
Rotation Axis X/Y/Z	jitterRotAxisRate	X,Y,Z

6. Interpolation

Observed UI choices include Linear, Ease-In-Out, Ease-In-Out (low), Ease-In, Ease-In (low), Ease-Out, and Ease-Out (low). Saved JSON has at least shown Linear, EaseInOut, and EaseIn in use. The UI label and stored string can differ. The exact interpolation curves used internally by VRM Live Viewer and the specific strength of (low) have not been verified.

7. Combining camera parts

One camera effect does not necessarily equal one datas entry. When moving a multi-entry part to another point in the timeline, preserve the internal startFrame offsets.

```
new startFrame = requested recipe start + (original startFrame - first startFrame of the part)
```

Do not recalculate the internal spacing from Changing Time; preserve the startFrame spacing of the referenced part.

8. 360-degree rotations

Actual testing confirmed that a 360-degree rotation can be specified with a single camera-data entry. For a one-way 360-degree rotation, use rotationHalf=True. With rotationHalf=False, the motion is back-and-forth: after reaching the

maximum angle, it reverses toward the starting side.

```
Verified example: enableRotationChange=True / rotationMode=MinMax / rotationHalf=True / rotationMin=0,0,0 / rotationMax=0,360,0 / rotationTime=24 / rotationTimingShift=0
```

A 360-degree rotation with rotationTime=24 was verified to take 12 seconds. The value remained 24 and the duration remained 12 seconds at both 30fps and 60fps. By contrast, startFrame and freezeFrame are frame-based: 12 seconds equals 360 frames at 30fps and 720 frames at 60fps.

If the end of a rotation does not align with the next startFrame, the rotation may be cut off early or appear to stop before the next camera begins. For timed effects, keep rotationTime consistent with fps and startFrame / freezeFrame.

9. Natural-language camera terms

Term	Meaning in this guide
Horizontal 360° orbit	Character-centered horizontal full orbit
Roll	Rotate the screen around the viewing axis
Dutch angle	Static tilted composition, e.g. 45°
90° roll / sideways composition	Character appears to lie sideways
Dolly / truck	Camera position movement using Changing Position
Jitter	Handheld-like or intense shake

10. Refinement guidance

Request	UI	JSON
Tilt too strong	Rotation Z	angle Z
Zoom too strong	Distance / Add Distance	distance / distanceMax
Rotation speed	Changing Rotation → Changing Time	rotationTime
Shake too strong	Jitter Amount	jitterPosAmount / jitterRotAmount
Sideways move too large	Changing Position → Add X	positionMax X

11. Output rules

- Preserve keys[n] ↔ values[n], order, and element counts.
- For the final VRM Live Viewer file, output one-line minified JSON.
- Pretty-printed JSON may be used for explanation, but provide a minified final file as well.
- Preserve strings such as True/False exactly as required by the saved format.

12. Rules for generative AI

- Prefer a close example from the supplied sample JSON instead of inventing unrelated values.
- Break compound directions into Position, Rotation, Distance, VAngle, and Jitter as needed.
- A 360° rotation can be specified in a single data entry. Use rotationHalf=True for one-way rotation.
- For timed effects, keep rotationTime consistent with fps and startFrame / freezeFrame.
- Preserve relative startFrame offsets inside multi-entry parts.
- When a user asks for refinement, identify both the UI control and JSON key whenever possible.
- Do not claim exact semantics for mappings marked as not established.

13. Example JSON

samples/full_camera_sample_6660f_160bpm.json is an example output combining multiple camera effects. It is not a preset for a particular song. Its frame placement uses an example condition of 160 BPM, 30 fps, start frame 0, and a total length of 6660 frames.

Terminology used in the public edition

Term	Meaning
Camera work	The overall camera movement, composition, and visual direction.
Frame	A position on the timeline, e.g. starting at frame 1800.
Camera data	One element of <code>datas[]</code> in an AutoCamera JSON file.
Camera part	A reusable camera setting made of one or more camera-data entries.

Unofficial status

This specification is unofficial documentation created by MoonaWorks. It is not an official specification from the developer of VRM Live Viewer. Please do not send questions about this project to the developer of VRM Live Viewer.